

The Job Sequencing and Tool Switching Problem

Part X: Position-Indexed Formulations

Two new exact formulations with $O(n|T|)$ constraints

Scope and status. This note introduces two *standalone* exact MILP formulations of the uniform SSP, both with constraint sets of size $O(n|T|)$ — polynomial in the input, fixed in advance — while the (exponentially many) configuration variables may be handled by column generation *without ever adding or modifying a constraint*. They are, to our knowledge, the first exact configuration-based SSP formulations with this property; all known alternatives carry per-configuration ordering constraints (MTZ, flow, SECs). **Proved:** exactness of both formulations; the plain LP relaxation of the first is always 0; tool-counting rows lift it to $\geq |U| - b$. **Verified:** the second formulation's LP can *strictly exceed* $|U| - b$ (the SSPMF root bound) and is tight on the 6-ring. Solution paradigms (CG, Benders) and open problems close the note.

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Motivation: Ordering Without Ordering Constraints

Jobs $J = \{1, \dots, n\}$ with tool sets $T_j \subseteq T$, $|T_j| \leq b \leq |T|$; $U = \bigcup_j T_j$; configurations $V = \{C \subseteq T : |C| = b\}$; $H_j = \{C \in V : T_j \subseteq C\}$; transition cost $d(C, C') = |C' \setminus C|$. Free-initial-load convention: Z_{SSP}^* counts insertions after a free first configuration.

Configuration-based formulations of the SSP must order the visited configurations. Every classical ordering device prices this per *configuration*: SECs are exponentially many; MTZ needs a potential and rows for each configuration pair; single-commodity flow needs a balance row per configuration. Since $|V|$ is exponential, all of these tie the *constraint* set to the configuration set — the obstruction identified in Part IV.

Position indexing (Shankar's proposal) removes the obstruction: time is discretised into positions $k = 1, \dots, n$, which are ordered *a priori*, so no ordering constraints exist at all. The price is that costs and feasibility must be expressed through position-indexed variables; the two formulations below differ in how they do this. Both rest on the following normal form.

Lemma 1.1 (Trajectory normal form). *Every feasible instance has an optimal schedule whose magazine trajectory consists of $m \leq n$ pairwise distinct configurations $C_{(1)}, \dots, C_{(m)}$, each serving at least one job, with cost $\sum_{l < m} d(C_{(l)}, C_{(l+1)})$.*

Proof. Take any optimal schedule with full magazines (pad arbitrarily; standard). Its trajectory has one configuration per job position. Collapse consecutive equal configurations (zero cost change). If a configuration recurs non-consecutively, reassign the jobs of the later visit to the earlier one (identical configuration) and splice it out: by the triangle inequality $d(X, Y) \leq d(X, C) + d(C, Y)$ (valid since $Y \setminus X \subseteq (Y \setminus C) \cup (C \setminus X)$ for $|X| = |C| = |Y| = b$) the cost does not increase. Delete job-less configurations by the same splice. Each step shortens the sequence, so the process terminates; what remains has $m \leq n$ since each surviving configuration serves a job. $\square$

Formulation PCF: Position–Configuration

PCF. Variables: $y_C^k \in \{0, 1\}$ ($C \in V$, $k = 1, \dots, n$): configuration C occupies position k ; $w_t^k \geq 0$ ($t \in T$, $k = 2, \dots, n$): insertion of tool t at position k . Abbreviate $a_t^k := \sum_{C \ni t} y_C^k$.

$$\begin{aligned} \min \quad & \sum_{t \in T} \sum_{k=2}^n w_t^k \\ \text{s.t.} \quad & \sum_{C \in V} y_C^k = 1 & k = 1, \dots, n \end{aligned} \tag{1}$$

$$\sum_{k=1}^n \sum_{C \in H_j} y_C^k \geq 1 \quad j \in J \tag{2}$$

$$w_t^k \geq a_t^k - a_t^{k-1} \quad t \in T, k = 2, \dots, n \tag{3}$$

$$\sum_{k=2}^n w_t^k \geq 1 - a_t^1 \quad t \in U \tag{4}$$

Constraints: $2n + (n-1)|T| + |U| = O(n|T|)$. Variables: $n|V| + (n-1)|T|$.

The model says: one configuration per position (repeats allowed and free); every job is covered

at some position; an insertion of t is charged whenever the presence mass of t increases from one position to the next; every used tool absent from the first configuration is inserted at least once.

Theorem 2.1 (PCF is exact). *The optimal value of PCF with binary y equals Z_{SSP}^* .*

Proof. ($\leq$) Take an optimal normal-form schedule (lemma 1.1). Set $y_{C_{(l)}}^l = 1$ for $l \leq m$ and $y_{C_{(m)}}^k = 1$ for $k > m$; set $w_t^k = \max(0, a_t^k - a_t^{k-1})$. (1), (2) hold by construction. At binary y , $a_t^k \in \{0, 1\}$ and $\sum_t \max(0, a_t^k - a_t^{k-1}) = d$ between the consecutive configurations, so the objective equals the schedule cost; repeats contribute 0. (4): a tool $t \in U$ with $a_t^1 = 0$ is required at some covered position, hence its presence rises from 0 to 1 somewhere and $\sum_k w_t^k \geq 1$.

($\geq$) An integer-feasible solution reads off a trajectory $D_1, \dots, D_n$ (one configuration per position). Its insertions at step k are exactly the tools with $a_t^k - a_t^{k-1} = 1$, each charged by (3), so the objective is at least $\sum_k d(D_{k-1}, D_k)$. By (2) every job is covered by some D_k ; assigning each job to a covering position and ordering jobs by position yields a feasible schedule of cost $\leq$ the objective. Hence the value is $\geq Z_{\text{SSP}}^*$. $\square$

Proposition 2.2 (Plain LP value is zero). *Without (4), the LP relaxation of PCF is 0 for every feasible instance.*

Proof. Pick $C_j \in H_j$ for each j and set $y^k = \frac{1}{n} \sum_j \delta_{C_j}$ for every k , $w = 0$. (1) holds; (2) gives $n \cdot \frac{1}{n} = 1$; a_t^k is constant in k so (3) is slack at $w = 0$. (The position-symmetric analogue of the Tang–Denardo LP-zero pathology: blending one mixture across all positions makes insertions, which are *differences*, invisible.) $\square$

Proposition 2.3 (Tool-counting rows are valid and give $\text{LP} \geq |U| - b$). *Rows (4) hold at every integer solution, and the LP relaxation of the full PCF is $\geq |U| - b$.*

Proof. Validity is the last step of the proof of theorem 2.1. For the bound, sum (4) over $t \in U$ and use $\sum_t a_t^1 = \sum_C |C| y_C^1 = b$ from (1): $\sum_{t,k} w_t^k \geq |U| - \sum_{t \in U} a_t^1 \geq |U| - b$. $\square$

Verified (verify_posform.py): IP = Z_{SSP}^* on the 6-ring, 5-ring and random instances; plain LP = 0.00 throughout; LP with (4) = $|U| - b$ *exactly* on every tested instance — the counting rows are the entire strength of the PCF relaxation so far. PCF thus matches the SSPMF root bound with a configuration-variable formulation.

Formulation PTF: Position–Transition

The PCF objective sees only per-tool *differences*; exact pair costs are recovered by indexing variables by *transitions* (Shankar’s $y_{i,j}^k$). Let $\perp \notin V$ be an absorbing terminal symbol and $A = \{(C, C') : C \neq C' \in V\} \cup \{(C, \perp) : C \in V\} \cup \{(\perp, \perp)\}$ (no diagonal pairs (C, C) — this is essential).

PTF. Variables $z_{(C,C')}^k \in \{0, 1\}$ for $(C, C') \in A$, $k = 1, \dots, n-1$: the k -th step moves from C to C' ($C' = \perp$: the trajectory has ended). Costs $d(C, C')$; $d(C, \perp) = d(\perp, \perp) = 0$. Write $\text{hd}_t^k = \sum_{(C,C') \in A: t \in C'} z_{(C,C')}^k$, $\text{tl}_t^k = \sum_{(C,C') \in A: t \in C} z_{(C,C')}^k$ (with $t \in \perp$ false), and $\text{ins}_t^k = \sum_{(C,C') \in A: t \in C' \setminus C} z_{(C,C')}^k$.

$$\min \sum_{k=1}^{n-1} \sum_{(C,C') \in A} d(C, C') z_{(C,C')}^k$$

$$\text{s.t.} \quad \sum_{(C,C') \in A} z_{(C,C')}^k = 1 \quad k = 1, \dots, n-1 \quad (5)$$

$$\text{hd}_t^k = \text{tl}_t^{k+1} \quad t \in T, k \leq n-2 \quad (6)$$

$$\sum_{(C,C') \in A: C' = \perp} z_{(C,C')}^k \leq z_{(\perp, \perp)}^{k+1} \quad k \leq n-2 \quad (7)$$

$$\sum_{(C,C') \in A: T_j \subseteq C} z^1 + \sum_k \sum_{(C,C') \in A: T_j \subseteq C'} z^k \geq 1 \quad j \in J \quad (8)$$

$$\sum_{k=1}^{n-1} \text{ins}_t^k \geq 1 - \text{tl}_t^1 \quad t \in U \quad (9)$$

Constraints: $(n-1) + (n-2)|T| + (n-2) + n + |U| = O(n|T|)$. Variables: $O(n|V|^2)$.

(6) makes consecutive steps chain (the tools of step k 's head are the tools of step $k+1$'s tail — aggregated *per tool*, which keeps the rows polynomial); (7) makes $\perp$ absorbing; (8) covers each job at the first tail or any head; (9) are the counting rows. Throughout assume $n \geq 2$ (for $n = 1$ the SSP is trivial and PTF has no steps). At integer points (6) alone already forces absorption ($\perp$ has no tools, and a configuration *is* its tool set); (7) is kept because it tightens the LP. Because diagonal pairs are excluded, a step either advances to a *different* configuration (cost ≥ 1) or terminates — blending cannot hide in free self-loops.

Theorem 3.1 (PTF is exact). *The optimal value of PTF with binary z equals Z_{SSP}^* .*

Proof. ($\leq$) Embed a normal-form trajectory (lemma 1.1): set $z_{(C_{(l)}, C_{(l+1)})}^l = 1$ for $l < m$, $z^{m \dots n-1}$ on $(C_{(m)}, \perp)$ then $(\perp, \perp)$. All rows hold (consecutive distinctness of the normal form is what the diagonal-free A requires); the cost telescopes to the schedule cost; (9) holds as in theorem 2.1.

($\geq$) At an integer solution, exactly one pair is chosen per step. By (6) the head tool-vector of step k equals the tail tool-vector of step $k+1$; since a configuration is its tool set, the chosen pairs chain into a trajectory $D_1 \rightarrow D_2 \rightarrow \dots$ until absorption, after which (7) keeps all later steps at $\perp$. The objective is exactly $\sum d(D_l, D_{l+1})$. By (8) every job is covered by a visited configuration; assign and order as in theorem 2.1. Hence the value is $\geq Z_{\text{SSP}}^*$. $\square$

Verified (verify_posform_f2.py): IP = Z_{SSP}^* ; and the LP relaxation is strictly stronger than PCF's:

instance	Z_{SSP}^*	LP(PTF)	LP(PTF, no (9))	LP(PCF)	$ U - b$
6-ring ($b=3$)	3	3.00	2.00	3.00	3
above-bounds ($b=3$, $Z_{\text{SSP}}^* > \max$)	3	2.10	2.05	2.00	2

On the 6-ring the PTF LP is *tight*; on an instance whose optimum strictly exceeds $\max(K^*-1, |U| - b)$, the PTF LP **strictly exceeds** $|U| - b$ — something neither PCF nor (at the root) SSPMF’s bound provides. Even without the counting rows the PTF LP is strictly positive (no Tang–Denardo collapse): with diagonals allowed it would be 0 (verified) — excluding them is what buys the strength.

Remark 3.2 (Relation to the hybrid proposal $y_C^k + x_{C,C'}$). The natural hybrid keeps PCF’s y_C^k and adds configuration-arc variables $x_{C,C'}$ carrying the cost, linked by $x_{C,C'} \geq y_C^k + y_{C'}^{k+1} - 1$. Exact linking requires one row per (pair, position) — the row set again grows with the generated configurations, violating the design criterion; aggregating the linking per tool reproduces exactly (6). In this precise sense **PTF is the hybrid formulation**: $z_{(C,C')}^k$ realises $x_{C,C'}$ position-wise, and the tails/heads play the role of y_C^k .

Can Per-Configuration MTZ Be Row-Reduced?

Shankar asked whether the per-configuration MTZ of Part IV can instead be thinned to polynomially many rows. The obstacle is structural: MTZ rows $v_C - v_{C'} + Nx_{C,C'} \leq N - 1$ are indexed by *arc variables*, and any ordering certificate (potentials, single-commodity flow balances, GG-flows) needs at least one row per configuration that can appear — the certificate must “know” each node. Aggregating potentials over clusters of configurations is exactly the cluster-MTZ of Part IV, which is provably not exact. Position indexing sidesteps the issue by making order *structural* rather than certified; we therefore regard PCF/PTF as the row-reduction of per-configuration MTZ, and pose the converse as section 5: a proof that no exact configuration-arc formulation has $o(|V_{\text{used}}|)$ ordering rows.

Comparison and Solution Paradigms

formulation	rows	IP exact	root LP	columns
per-config MTZ (Part IV)	grows with V_{gen}	yes	untested	$O(V ^2)$
cluster MTZ (Part IV)	$O(n^2)$	no	weak	$O(V ^2)$
PCF	$O(n T)$, fixed	yes	$= U - b$ (verified)	$n V $
PTF	$O(n T)$, fixed	yes	$\geq$ PCF; $> U - b$ possible	$O(n V ^2)$

Solution paradigms (consequences, not design drivers). *Column generation*: in both formulations a generated configuration (PCF) or configuration pair (PTF) contributes columns to existing rows only; pricing is a set-selection problem over b tools (PCF: single set with job-coverage bonuses, the JGP pricing class; PTF: a coupled pair — harder, and the practical price of its stronger LP). *Benders*: in PTF one may project the z -layer onto a surrogate θ per position and cut from the per-tool consistency duals; whether this beats direct CG is untested. *Hybrids with the job-level BBC* (Part VIII) are an implementation topic deliberately kept out of this note.

OP-X1. Characterise the PTF LP bound; in particular prove (or refute) $\text{LP}(\text{PCF}) \leq \text{LP}(\text{PTF})$ in general, quantify how far past $|U| - b$ the PTF LP can go (it reached Z_{SSP}^* on the ring), and compare against the Catanzaro F3/F4 root bounds on Tabela1C.

OP-X2. Polynomial row families beyond the counting rows: position-conflict cliques, per-tool interval/spread rows tying w_t to the positions of jobs needing t , and symmetry breaking (reversal + position shifts); which are facet-defining for the PCF/PTF polytopes on small instances (PORTA)?

OP-X3. Practical CG for PTF: pair pricing as a quadratic set-selection (two coupled b -subsets with d -cost and dual rewards) — exact MILP pricing vs. heuristic pricing with exact fallback; degeneracy management for the $n - 1$ identical step rows.

OP-X4. Lower bound for arc-based ordering: prove that every exact formulation of the SSP whose integer solutions are characterised by configuration-arc incidence requires $\Omega(|V_{\text{used}}|)$ constraints, or exhibit a sub-linear ordering certificate (this would also row-reduce per-config MTZ directly, Shankar's original question).

References